

HPIM — Human Proximity Interpretation Module

Parent Standard: Physical Reality Interpretation Layer (PRIL™)

Category: AI & Interpretation

Subcategory: Human Proximity Interpretation

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Compatibility: OOF® Methodology OS™ · PRIL™ · CLIA® · RIS · INTEGROS® · EVIP® · SIMULOS®

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Canonical Definition

Human Proximity Interpretation Module defines the methodological conditions under which autonomous systems interpret human presence, distance, approach, crossing, unsafe closeness, and proximity-related risk before generating physical action or embodied response.

A system satisfies HPIM only if:

- human presence is interpreted before physical execution
- proximity is evaluated under defined conditions
- approach, crossing, and nearby movement are recognized
- unsafe closeness is treated as an execution restriction condition
- physical action is limited when human proximity interpretation is insufficient

A system that detects humans without governed proximity interpretation does not satisfy HPIM.

Module Function

HPIM defines the human-nearby interpretation layer of embodied execution.

It ensures that human detection alone is not treated as sufficient for safe physical operation.

The module applies wherever systems operate near humans under conditions in which distance, movement, closeness, crossing, or contact risk affect execution validity.

Minimum Implementation Framework (MIF)

Step 1 — Define the Human Proximity Boundary

The organization must define where human proximity becomes operationally relevant.

Minimum requirement:

- the proximity boundary is explicit
 - human-nearby conditions are distinguishable
 - undefined human presence is excluded from valid execution logic
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Step 2 — Define Human Presence Interpretation

The system must define how human presence is interpreted.

Minimum requirement:

- human presence is not treated as raw signal only
 - interpretation conditions are explicit
 - uncertain human detection is not silently treated as valid certainty
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Step 3 — Define Distance and Approach Logic

The system must define how distance and movement toward the system are interpreted.

Minimum requirement:

- distance thresholds are explicit
 - approach, crossing, and surrounding movement are distinguishable
 - unsafe closeness conditions are identifiable
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Step 4 — Define Restriction Conditions

The system must define when execution must slow, stop, restrict, or escalate due to human proximity conditions.

Minimum requirement:

- restriction logic is explicit
 - proximity uncertainty can restrict execution
 - continued action under unsafe closeness is not treated as valid
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Step 5 — Preserve Human Priority

The system must preserve human safety above speed, efficiency, throughput, or task completion.

Minimum requirement:

- human proximity risk overrides operational convenience
 - execution does not continue because the task is urgent
 - physical safety remains structurally superior to performance
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Step 6 — Restrict Invalid Execution

The system must not be treated as valid if physical execution proceeds while human proximity interpretation is absent, degraded, or unreliable.

Minimum requirement:

- invalid execution conditions are identifiable
 - degraded proximity interpretation blocks valid reliance
 - sensing alone does not restore interpretation validity
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Use Case 1 — Collaborative Industrial Robot

Scenario

A robotic arm operates in a shared workspace where humans may approach, cross nearby, or enter a hazardous movement zone during operation.

Application

HPIM is used to ensure:

- human proximity boundaries are defined
 - presence, distance, and approach are interpreted
 - unsafe closeness triggers restriction
 - task execution does not override human safety
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Result

The organization gains a stronger basis for preventing robotic operation from continuing when human-nearby conditions are not sufficiently interpreted.

Use Case 2 — Humanoid or Mobile Service Robot

Scenario

A mobile or humanoid system operates in an environment where humans move unpredictably around it and physical response depends on reliable

interpretation of nearness and motion.

Application

HPIM is used to ensure:

- nearby human conditions are interpreted before movement
 - crossing, approach, and unstable proximity are recognized
 - execution slows, stops, or escalates when proximity certainty is insufficient
 - physical action is bounded by human-priority conditions
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Result

The organization gains a structurally governed human-nearby interpretation layer instead of relying on raw sensing or assumed safety.

Canonical Closing Statement

If human proximity is not interpreted before action, physical execution is not structurally safe.

Related Documents

→ [Physical Reality Interpretation Layer \(PRIL™\)](#)

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Canonical Source

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